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3. Agenda of the Institute Sustainability Report 2022-23

We, at the Sustainability Cell, IIT Bombay, are immensely proud to bring to you the Institute's first-ever sustainability report. As the world grapples with complex challenges like global warming and climate change, we are committed to fostering a culture of sustainability and shared responsibility at IIT Bombay. The **Institute Sustainability Report** is the culmination of a months-long effort and is a testament to our commitment to creating a healthier planet and better living conditions for all its inhabitants.

The report will begin with a brief introduction to IIT Bombay, followed by a detailed catalogue of our student activities, research, and policy interventions categorized under the United Nations' Sustainable Development Goals (SDGs). The UN SDGs are a set of 17 global goals adopted by the United Nations in 2015 as a universal call to action to end poverty, protect the planet, and ensure that all people enjoy peace and prosperity by 2030. At the end of the report, we have also included a bonus section on biodiversity!

Of course, we cannot enlist every contribution from our dedicated students, faculty members, and staff, but we have tried our level best to capture the essence of sustainability at IIT Bombay. We hope that this report is the first of many and serves as a benchmark for our future efforts.

IIT Bombay secured first place in India and is ranked in the range 281-300 globally in the QS Sustainability Ranking 2023. The Institute is ranked 173rd in the 'Environmental Impact Performance' and 484th rank in the 'Social Impact Performance' category of the QS Sustainability Ranking 2023. The Institute was awarded a silver medal in the 'Environmental Impact' category for the productivity of research into the United Nations' Sustainable Development Goals (SDGs) by QS.

About the Sustainability Cell

IIT Bombay's Sustainability Cell is a student-run body fostering sustainable change at IIT Bombay. We aim to achieve this by promoting behavioural, technological, and policy changes in the institute. We are raising awareness about sustainability through student engagement via events & competitions, ushering in efficient technologies like motion sensor lighting & waterless urinals, and collaborating with other institute bodies, staff, and students to achieve our goals.

4. The Director's Message

5. IIT Bombay at a glance

IIT Bombay (Indian Institute of Technology Bombay) is an Indian public engineering and technology institution in Mumbai, India. Founded in 1958, it is one of the most prestigious engineering institutes in India and is consistently ranked among the top institutes in the country. The campus of IIT Bombay is situated in the Powai area of Mumbai, spread over 550 acres. It is surrounded by lush green hills, lakes, and the Sanjay Gandhi National Park, offering a natural and serene environment for its diverse cohort of 13,000+ students.

The institute hosts 15 academic departments, two schools, four interdisciplinary programs, and thirty-seven centres that offer quality and holistic education. Several of these focus on sustainability, including the Environmental Science and Engineering Department, the Centre for Technology Alternatives for Rural Areas, and the Department of Energy Science and Engineering. More than 680 faculty members who are not only among the top in the nation but also highly regarded internationally for accomplishments in education and research serve the institution.

Sustainability at IIT Bombay has been a prime area of interest since its establishment. Through the years, the institute has demonstrated a solid commitment to creating and balancing a comfortable campus along with preserving the environment. We have made significant efforts towards sustainability in various areas like energy, water, waste management, buildings, transportation, and education, and remain committed to reducing our environmental impact and promoting sustainable practices on campus.

Additionally, there are numerous student-led initiatives promoting sustainability at IIT Bombay. These include the Sustainability Cell, Abhyuday (social welfare), Team Zero Waste (sustainable and scalable waste management), Team Shunya (affordable, net zero housing), Energy Club (spreading awareness about renewables), and National Service Scheme (educational outreach, environmental conservation, social development).

6. The United Nations Sustainable Development Goals (UN SDGs)

The UN SDGs are a useful and standardised tool for assessing and reporting on sustainability efforts. In the following section, we catalogue our activities, research, and policy interventions under each SDG to identify areas where we are making a positive impact, as well as areas where more work is needed. By aligning our sustainability efforts with the SDGs, we hope to ensure that we are making a meaningful and measurable impact on the global sustainable development agenda.

SDG 1: No Poverty

- **Equality of opportunity: admissions reservations, fee remittances, and scholarships**

The institute is committed to providing equal opportunities to all students, regardless of

their financial background. Since the 2019-2020 admissions cycle, the institute has been reserving 10% of seats for students belonging to Economically Weaker Sections (EWS). Additionally, the institute offers fee remittances for underprivileged students to ensure they have equal opportunities to succeed and do not have to worry about making ends meet to pursue an education. Furthermore, the institute provides merit cum means scholarships and free messing facilities to academically gifted students to help cover the cost of tuition, books, and other expenses. During the academic year 2021-22, about 600 scholarships worth more than ₹300 million were granted, some prominent ones being Desai Sethi, IIT Bombay HF Named scholarships, Round Memorial, C'1990, C'1992 and Indira Foundation scholarships. The institute is dedicated to creating a diverse and inclusive community where all students have the resources they need to thrive.

- **Providing laptops and broadband support during the pandemic and beyond**

The institute provides laptops and other necessary technological devices to financially disadvantaged students to ensure they can access the tools needed for their studies. This initiative began during the COVID-19 pandemic when the institute's functioning had to be shifted online, leaving students with no access to computers at libraries or laboratories. Demonstrating its commitment to bridging the digital divide caused by varying financial backgrounds, the institute crowdsourced ₹4 crores from its alumni and the general public during the pandemic. Additionally, the IIT Bombay Heritage Fund (IITBHF) raised ₹2.35 crore to purchase 500 laptops and broadband support for 500 needy students. This was done to ensure that all students could access the same content and media while working on practical projects with ease.

- **Affordable accommodation for students**

Mumbai is one of the most expensive cities in India. The institute is located in a relatively higher profile area in the city, making it harder for students to find inexpensive residences near campus. The institute has addressed this issue by providing extremely affordable housing in hostels. The institute also provides fee remittances to cover housing expenses for financially disadvantaged students. New and bigger hostels are being built to accommodate more students and should be ready by 2025. The hostels have various amenities such as games rooms, gyms, study rooms, music rooms, dance rooms and TV rooms.

- **Significantly cheaper tuition costs compared to top universities abroad**

Despite having an exceptional employability track record and being globally renowned as a leader in quality education, IIT Bombay offers extremely affordable baseline tuition costs (even without scholarships) compared to top international universities. As an example, it is an impressive 20-25 times cheaper than Stanford! (considering full tuition and no scholarships). The institute provides top-notch facilities, experienced faculty, and

valuable opportunities for students at a fraction of the cost of overseas institutions. This is a testament to the commitment of the institute to provide world-class curricula, cutting-edge research, and a supportive student community to talented individuals, regardless of their financial background.

- **Activities of Abhyuday - the social body of IIT Bombay**

Abhyuday's involvement in quality education for all classes of students has been commendable. Guidance sessions detailing possible career opportunities and the importance of choosing the right career trajectory were conducted for about 30,000 students from 500+ schools, especially underprivileged students from government schools, under the 'Career Counselling Campaign' (CCC). As an extension of CCC, Abhyuday launched the 'Abhyuday Helpline App' in 2017, comprising a network of 140+ experts from various premium institutes across India to resolve issues pertaining to studies and careers faced by students in rural areas. Furthermore, they have conducted 2 successful Video Tele-Conferences (VTCs) for BMC schools and guided students over 100+ calls on toll-free numbers. Abhyuday also organised 20+ 'Masti Ki Paathshalas' (MKPs) in various NGOs that promote learning science with innovative and fun pedagogies.

SDG 2: Zero Hunger

- **Abhyuday's food distribution campaigns**

Abhyuday, the social body of IIT Bombay, is dedicated to tackling the problem of food insecurity. Through ten successful food distribution campaigns, Abhyuday has provided over 2000 meals to those in need. The campaigns were made possible with the help of over 150 volunteers from within and outside the institute who generously donated resources and time to support the cause. With plans to continue organising food distribution drives, Abhyuday is working to make a lasting impact in the local community.

- **Subsidised mess rates**

The institute has made sure to provide nutritious and wholesome food at extremely affordable rates in its various hostel messes. The food offered is comparable to home-cooked meals in terms of nutritive value and is available at a subsidised price, ensuring that students have access to healthy and nourishing food. This initiative not only benefits the students but also promotes a healthy lifestyle. The institute's commitment to providing quality food at an economical rate is a testament to its focus on the well-being of its students.

- **Activities at CTARA**

The Centre for Technology Alternatives for Rural Areas (CTARA) organised an Institute

Talk on the "Potential of Millets - From Farm to Plate", which discussed the benefits of incorporating millets into our diet and their role in addressing food security and malnutrition. The talk emphasised the need to shift towards traditional, locally available food grains such as millets and highlighted their potential in creating a sustainable food system. The event aimed to create awareness and promote the consumption of millets, which are considered to be nutritionally superior to other staple crops. The talk was attended by students, faculty, and members of the local community, who actively participated in the discussions.

Additionally, CTARA's Nutrition Group focuses on promoting health and nutrition among students. The group provides workshops, counselling, and outreach activities to educate students on the importance of healthy eating habits and lifestyle choices. They aim to create a healthier campus community and address issues such as obesity, malnutrition, and dietary problems through their various initiatives.

- **Workshop on sustainable agriculture**

The Sustainable Development Policy Study Unit of the Centre for Policy Studies held a workshop on Sustainable Agriculture. The workshop aimed to raise awareness and educate participants on the importance of sustainable agriculture practices. Experts and practitioners shared their knowledge and experiences, including discussing policy incongruence, risk and returns in sustainable agriculture, and mainstreaming sustainability in agriculture in Maharashtra. The workshop provided valuable insights and information on the topic and encouraged participants to incorporate sustainable agriculture practices into their lives.

SDG 3: Good Health & Well-being

- **IIT Bombay hospital services – free for students, staff and faculty**

The institute has its own hospital that provides comprehensive medical services to students, staff, and faculty. The hospital is equipped with state-of-the-art facilities and staffed by experienced medical professionals. The services provided include general medical consultations, diagnostic tests, and treatment for minor illnesses and injuries. The hospital also has specialised clinics for specific medical needs, such as dentistry, ophthalmology, and nutrition. The hospital services are free for students, staff, and faculty members, ensuring that access to medical care is not a barrier to students' success. Moreover, IIT Bombay students, staff, and faculty are entitled to a 20% discount in all medical services at the Hiranandani Hospital, one of Mumbai's foremost multispeciality hospitals.

- **Student Mentorship Program (SMP)**

The institute has a robust student mentorship program that helps first-year

undergraduates adjust to institute life and addresses any problems they may face. The program pairs a group of freshers with a senior who serves as a mentor. The mentor is a trained student leader who serves as a guide and a resource for the freshers throughout their first year at the institute. The mentor helps the freshers navigate the resources available on campus, provides academic and social support, and serves as a sounding board for any concerns or issues the freshers may have. The program aims to make the transition to institute life as smooth as possible for the freshers and to help them achieve their academic and personal goals.

- **Student Wellness Center (SWC)**

The institute has a Student Wellness Center, which provides free mental health counselling to help students cope with stress, anxiety, depression, and other mental health issues. The centre also conducts regular talks and seminars on mental health and well-being, to educate and raise awareness among students. The centre is staffed by trained professionals who are dedicated to helping students navigate the challenges of college life and maintain good mental health.

- **Yogastha, an NSO initiative aimed at general well-being**

Yogastha is an initiative of the institute's National Service Organization (NSO) aimed at promoting general well-being among students. The program includes yoga and meditation sessions, which are held regularly to help students reduce stress, improve focus, and enhance overall physical and mental health. The program is open to all students and is designed to cater to the needs of beginners as well as more experienced practitioners. The initiative also conducts workshops on various aspects of Yoga and meditation to help students to understand the benefits and techniques of the practice.

- **IDDDP in Digital Healthcare**

The institute offers an innovative interdisciplinary dual degree program in digital healthcare. This program combines the strengths of multiple disciplines to provide students with an in-depth understanding of the intersection of healthcare and technology. This program features coursework in computer science, healthcare, and business, with a focus on the development and application of digital technologies in healthcare. The program includes hands-on experience through internships, research projects, and capstone projects, allowing students to work on real-world problems and make an impact in the field. The IDDDP is one-of-its-kind in India, designed to prepare students for careers in a rapidly growing field and to be future leaders in digital healthcare.

- **Cup for a Change**

The institute's NSS unit hosted an insightful session on menstrual cups, where Noureen Aysha, Co-Founder of FemiSafe, shared the benefits of these products for women's health and the environment. The event aimed to raise awareness and break the taboo surrounding menstrual health, and empower college students to make a positive impact in the world. During the session, attendees also participated in a fun period cramp simulation activity to understand the challenges faced by women during their periods, and why menstrual cups are a game-changer. The event was a huge success, and all attendees had a chance to receive a menstrual cup for free, as a token of appreciation for their participation. The session was a step towards a more sustainable and healthier future for women.

- **Consumption of narcotics and alcohol banned on campus**

The institute strictly prohibits the consumption or possession of narcotics or alcohol on campus. This policy applies to all students and is strictly enforced by the campus security and the student conduct office. Any violation of this policy will result in disciplinary action, including fines and/or expulsion from the institute. The institute promotes a drug-free and healthy environment and encourages students to make responsible and healthy choices that support their academic and personal success.

- **Abhyuday's Activities**

Abhyuday, the social body of IIT Bombay, undertook various activities like menstrual hygiene awareness workshops and blood donation drives to foster good health and well-being. On the occasion of International Women's day, Abhyuday visited 3 NGOs in association with Qause, conducted workshops on the importance of menstrual hygiene, and distributed 100+ sanitary pads. Moreover, as part of their blood donation drive, 316 units of blood, which have the potential to save 948 lives, were collected from students, faculty, staff, and other residents.

SDG 4: Quality Education

- **Consistently ranked among India's best universities**

The institute is consistently ranked as India's top university by prestigious ranking systems like QS World Rankings and NIRF Rankings. The QS World Rankings places the institute in the top 200 universities globally, and it is consistently in the top 3 colleges in the NIRF Rankings. These prestigious ranking systems evaluate universities based on various factors, including academic reputation, research output, and the employability of graduates. The institute is renowned for its academic excellence, world-class faculty, cutting-edge research, and state-of-the-art facilities. The institute's reputation for academic excellence continues to attract the best and brightest students, faculty, and staff from around the world, making it one of the most sought-after universities in India.

- **Established track record of admissions in top graduate programs abroad**

Every year, multiple students from the institute go to top institutions abroad for post-graduation studies. The institute's reputation for academic excellence, the quality of its students, and the broad range of programs it offers make it an attractive stepping stone for students seeking to pursue higher education abroad. The institute has a dedicated office that provides guidance and support to students applying to international universities and also offers a wide range of scholarships, grants and fellowships to support students financially. The institute also collaborates with foreign universities to organise research internships for students pursuing post-graduation studies abroad. The institute's students have been admitted to top institutions such as Harvard, Oxford, Cambridge, MIT, and many more.

- **Interdisciplinary academic programs with HASMED courses**

The institute offers a wide range of interdisciplinary academic programs that integrate courses in humanities, arts, social science, management, entrepreneurship, and design. These programs aim to provide students with a holistic education that combines technical skills and knowledge with critical thinking, creativity, and communication. Every undergraduate has to compulsorily undertake two HASMED courses in their first year and a few more later on to satisfy the requirements for their degree. The interdisciplinary approach of these programs allows students to explore diverse perspectives and gain a deeper understanding of the complex issues facing society today. The curriculum is designed to develop students' problem-solving abilities and creativity, encouraging them to think outside the box and tackle real-world problems. These programs prepare students for dynamic careers in a rapidly changing world.

- **Educational Outreach by the National Service Scheme (NSS)**

NSS, IIT Bombay, through its 'Open Learning Initiative' (OLI), aims to break language barriers in education throughout the country. OLI provides educational videos in 9 different languages through its YouTube channel, which has been active for the last seven years. The channel has reached out to 125K+ learners, including numerous students from Naxalite-affected regions. 'Dhruva', another commendable initiative, provides academic and motivational help to underprivileged kids for national-level competitive exams including the Joint Entrance Exam (JEE), National Talent Search Exam (NTSE) and the Indian National Olympiads. They have collaborated with an NGO based in a village in Bihar where they trained 40 students from financially weak backgrounds - out of them 19 made it to the IITs! They are also conducting a trial run with the Government of the state of Odisha for 500 students under this initiative. Through their 'BMC awareness drive' initiative, NSS volunteers visit various Bombay Municipal Corporation (BMC) schools to create awareness about various sustainable practices like rainwater harvesting, waste segregation, etc. among the budding minds of our nation. They have been creating awareness among ~5000 kids in BMC schools close to IIT Bombay. This process has been assisted by a YouTube channel, "Tarang", which has more than 115K views.

- **Activities of the Energy Club**

In partnership with the NGO 'Salaam Bombay', IIT Bombay's Energy community mentored a team of students from Classes 9 to 12 representing India at the FIRST Global Challenge (FGC) 2022, an Olympics-style international robotics competition held in Geneva. They guided the students on ideation, problem identification, designing solutions, and report-making, for their ideas on carbon capture and energy efficiency. Moreover, the Energy Club conducted an interactive workshop for 100 prospective teachers at Mahatma Education Society's Pillai College of Education and Research on Akshay Urja Diwas 2022. The workshop aimed to sensitise the teachers on the impact of consumerism on energy consumption and climate change due to food wastage, fast fashion, single-use plastics, and conventional transport.

SDG 5: Gender Equality

- **Supernumerary seats for female students**

The institute has implemented a policy of providing 20% supernumerary seats for female students to boost gender representation. This policy aims to increase the number of female students in the institute and create a more diverse and inclusive community. The institute believes diversity and gender representation are crucial for creating an environment that fosters learning, creativity and innovation. The supernumerary seats for females are essential to ensuring that the institute's student body is representative of the society it serves and promotes equality of opportunity.

- **Gender Cell**

The institute has a Gender Cell that works to promote and foster inclusivity and provide redressal for harassment. The Gender Cell is responsible for creating and maintaining a safe and inclusive environment for all members of the institute community, regardless of their gender. The Cell supports and assists students, staff, and faculty members who have experienced harassment or discrimination based on gender. The Gender Cell works closely with other organisations to develop and implement policies and programs that promote gender equality and inclusivity. It also conducts workshops, seminars, and other events to raise awareness about gender-related issues and to promote a culture of respect and understanding.

- **No discrimination policy**

The institute has a strict no-gender discrimination policy. The policy prohibits discrimination based on gender, including harassment and violence, in any form and under any circumstances. The policy applies to all members of the institute community and is enforced by the Gender Cell, which is responsible for investigating and addressing complaints of gender discrimination. The institute is committed to creating a safe and inclusive environment for all members of the institute community and promoting equality of opportunity for all students, staff, and faculty members, regardless of their gender. The institute encourages individuals who have experienced or witnessed gender discrimination to report it promptly, to ensure that appropriate action is taken.

- **GC101 – compulsory sensitisation course for all students for appropriate professional behaviour**

GC101 is a compulsory sensitisation course run by the Gender Cell for all students. The course aims to educate students on appropriate professional behaviour and promote gender sensitivity and inclusivity. The course covers a wide range of topics, such as understanding gender and its implications, preventing and addressing gender discrimination and harassment, and creating a culture of respect and inclusivity. The course is designed to help students understand the importance of gender sensitivity and to develop the skills necessary to interact effectively with people from diverse backgrounds. The course is mandatory for all students and is held at the beginning of their first year. This ensures that all students have a foundation of knowledge and skills to interact respectfully and inclusively.

- **Promoting women in STEM and other fields**

Many of the institute's faculty are female, promoting women in STEM and other fields. The institute values diversity and gender equality and is committed to creating an inclusive community for all members of the institute community. This not only promotes female representation in these fields but also helps create a more inclusive and equitable environment for all students. The institute encourages female students to pursue their interests and passions and provides them with the support and resources they need to succeed in their chosen fields. The institute serves as a role model for

women in STEM and other fields by having many female faculty members.

- **Saathi - the LGBTQ+ resource group**

Saathi is the LGBTQ+ resource group at the institute, working towards creating a supportive and inclusive environment for students and staff of diverse sexual orientations and gender identities. The group organises awareness campaigns and events to promote LGBTQ+ rights and educate the institute community on these important issues. Saathi also provides a safe space for individuals to express themselves and connect with like-minded individuals.

SDG 6: Clean Water and Sanitation

- **Clean and free water for all**

The provision of safe drinking water facilities in all residential and academic buildings in the institute is aligned with the goal of ensuring the availability of clean water and sanitation for all. It curbs the spread of waterborne diseases and consequently improves the living standards for everyone. Furthermore, the institute takes every step necessary to ensure that the water is supplied sustainably, thus conserving and protecting water resources. To cite a few, the institute organised an open-to-all brainstorming session wherein students came up with projects pertaining to rainwater harvesting system design, and identification of water-intensive activities and proposing solutions for the same. A team of 5-7 students was selected for each project, with project leads being senior students with experience in that particular field.

- **Integrated Wetland Technology**

The Integrated Wetland Technology (IWT) is an innovative natural treatment system for sewage, which was devised by the institute to overcome the constraints of conventional sewage treatment plants. These constraints, such as the need for a regular power supply and skilled manpower, limit the practical application of traditional systems.

The IWT is an engineered wetland system that utilizes wetland vegetation, support media, and their associated microbial assemblages to help treat wastewater. It consists of a primary, secondary, and tertiary treatment system. The Primary Solid Removal Tank (PSRT) acts as a primary treatment system and removes 60-70% of suspended solids along with 35-40% of Biological Oxygen Demand (BOD). The Planted Bed behaves as a secondary treatment system and reduces 80-90% of BOD. The two-stage sludge management system lengthens the frequency of sludge removal from the planted bed and provides an additional edge to IWT compared to most of the wetland-based technologies.

The final treated water from the system meets discharge standards set by the Central Pollution Control Board (CPCB) and can be used for secondary purposes such as irrigation, gardening, and toilet flushing. The innovative design of IWT makes it a technologically feasible and economically viable solution to treat wastewater in India and

worldwide. By utilizing natural processes for sewage treatment, the IWT conserves energy and fosters the transition to a low-carbon economy.

- **N-Treat: In-situ Nallah Treatment Technology**

Wastewater from households is carried by open nallahs and discharged into nearby water bodies, leading to pollution, harm to aquatic life, and health hazards for humans. To prevent this, treatment of nallah wastewater is necessary prior to discharge. However, availability of land space for setting up wastewater treatment units is a major challenge. To tackle this issue, the institute collaborated with EMERGY Enviro Private Limited to develop N-Treat, a technology that treats wastewater without requiring additional space. N-Treat uses various physical and biological processes to treat the wastewater within the nallahs. It has been deployed by the Brihanmumbai Municipal Corporation (BMC) to prevent sludge and sewage from 25 storm water drains between Bandra and Dahisar in Mumbai from flowing into the sea.

SDG 7: Affordable and Clean Energy

- **Research into cleaner energy**

Prof. Anand B Rao, the Head of the Department at the Centre for Technological Alternatives for Rural Areas (CTARA), researches renewables, energy efficiency, carbon capture, and climate change mitigation using technological interventions in rural areas. Currently, his group is working on projects like 'Impact assessment of large-scale renewable energy systems', 'dynamic life cycle sustainability assessment of mini grids', and the 'localisation of SDGs'.

Prof. Arnab Dutta, an Assistant Professor in the Chemistry Department and Associate Professor at the Interdisciplinary Programme in Climate Studies, researches green hydrogen, carbon capture and utilisation, and renewable energy. In the past, he has researched the production process of hydrogen from water and taught a Continuing Education Program on Carbon Capture, Utilization and Storage. Currently, he is involved in several projects including hydrogen production catalyst development, biomass to hydrogen conversion, carbon dioxide capture and utilisation, establishing a carbon dioxide capture pilot plant, and development of a water electrolyser prototype.

Prof. Chirag Deb is an Associate Professor at the Centre for Urban Science and Engineering. Prof. Deb's research focuses on impactful areas like Building energy modelling, Urban data Analytics and Renewable Energy Integration in Urban Contexts. His doctoral and graduate students are pursuing research on cost-optimal energy retrofit of buildings, urban energy optimization, urban heat island effects, optimization of energy and indoor environmental quality in buildings, and application of building-integrated photovoltaics.

Prof. Ojus Mohan of the Department of Chemical Engineering researches into the rational design of sustainable and thermodynamically stable catalytic materials as well as the nexus of energy, climate and sustainability. He leads the Catalysis Research for Sustainability Lab on campus. Currently, his group is pursuing research on carbon-dioxide conversion to hydrocarbon and methane and production of green hydrogen via methane cracking.

- **Energy Efficiency on Campus**

At IIT Bombay, every room has been fitted with efficient and modern electric equipment, allowing for greater energy efficiency and improving energy access. In an effort to promote sustainability and a low carbon footprint, the institute has sponsored a number of projects led by professors, such as the installation of motion sensors in common areas, waterless urinals, solar water heaters, perforated exam answer sheets for detachment and subsequent reuse of blank sheets, and timed remote switches for lab equipment, among others.

- **Ojas - the Energy Swaraj Club**

To celebrate 75 years of Independence, the Energy Club, IIT Bombay, launched Ojas - the Energy Swaraj Club. The initiative works towards realising Energy Swaraj in India, ensuring energy security and climate change mitigation under Prof. Chetan Solanki, the 'Solar Man of India'. Currently, the Energy Swaraj Club works on initiatives like hands-on Solar Lamp workshops on campus.

SDG 8: Decent Work and Economic growth

- **India's Most Employable Institute**

As per the QS Graduate Employability Rankings 2022, IIT Bombay is India's most employable institute, making impressive contributions to the country's economy. The institute is internationally acclaimed for its excellent academic programs, world-class faculty, cutting-edge research and state-of-the-art facilities. As the only Indian university with a 200 score on QS' 'Partnerships with Employers' indicator, the institute captures an appreciable number of research engagements and work-placement partnerships.

The institute extensively prepares its students for their future by providing them with an erudite understanding and access to resources. In the year 2021-22, IIT Bombay students received job offers from 332 companies, with a total of 1878 jobs offered. The institute attracted top recruiters like Google, Amazon, Microsoft, Goldman Sachs, Facebook, and many more. With a consistently increasing number of students placed every year, the institute boasted an exemplary total of 1441 students placed in the year 2021-2022. Furthermore, the number of internships offered has been on the rise as well, with 1195 internship offers in the past year.

- **Desai Sethi School of Entrepreneurship**

The Desai Sethi School of Entrepreneurship, DSSE of the institute trains aspiring entrepreneurs through courses and structured mentoring programs. Led by a team consisting of institute faculty, adjunct faculty, officers, and mentors, the school offers several courses and labs relevant to innovation and entrepreneurship open to both undergraduate and graduate students. All these courses are complemented by experiential learning in various aspects of product innovation and business lifecycle through mentoring programs. As a result of the school's efforts, more than 25 start-ups have been established by students exposed to entrepreneurship through DSSE courses.

- **Activities of the Entrepreneurship Cell**

The institute's Entrepreneurship Cell has held several major events to cultivate a strong culture of entrepreneurship among the students. The institute hosted the 18th edition of the Entrepreneurship Summit on January 28th and 29th. E-Summit '23 provided a platform for discussion on innovative ideas that have the potential to transform current challenges into opportunities for positive change. Furthermore, it conducted Eureka!, Asia's largest business model competition (accredited by CNN & Thomson Reuters). Eureka! provides a 360-degree holistic experience in entrepreneurship in the five-month period it spans. This event starts with acknowledging the potential of an idea, goes on to draft a B-model, and finally presents it to famous investors.

- **Technology Business Incubator (TBI)**

IIT Bombay's Centre for Policy Studies (CPS) hosts a technology business incubator, which has a wide spectrum of incubatee companies. Technology based new enterprises are high risk and high growth ventures, and require an enabling environment like the CPS-TBI to enhance the prospects of their success. It creates jobs, promotes new technology, knowledge, and innovation-based startups and provides a platform for speedy commercialization of technologies developed by the institute or by an individual.

- **Profectus by Abhyuday**

Profectus is a fellowship program organised by IIT Bombay's student-run organisation Abhyuday. The program provides a platform for students to connect with the social sector and contribute to change. Through the program, students are allowed to explore their interests and develop the skills required to make an impact in the social sector. The program consists of workshops, lectures, and activities that help students gain knowledge and hone their skills. Additionally, Profectus also provides mentorship and guidance that helps students make the most of their social impact journey.

SDG 9: Industry, Innovation & Infrastructure

- **SINE: Society for Innovation and Entrepreneurship**

The institute's SINE is an umbrella organisation for fostering entrepreneurship and nurturing tech start-ups. It administers a business incubator which provides start-to-scale

support for technology-based entrepreneurship and facilitates the conversion of research activity into entrepreneurial ventures. Set up in 2004, it is one of the earliest incubators in academia with the potential to create startups focussing on economic growth, strategic value and social relevance. It gives prototyping grants, provides incubation support to tech startups, and also runs accelerator programs in corporate partnership.

- **Maximising Computation in Blockchains**

Professors Umesh Bellur and Vinay J. Ribeiro from the Computer Science and Engineering department have proposed an on-chain protocol to enable computation-intensive transactions (CIT) on blockchains, such as Ethereum, without compromising security. Currently, blockchains allocate only a tiny fraction of the block inter-arrival time for validating smart contracts present in transactions, which causes forking and increases unfairness. Offloading validation to non-chain nodes suffers from trust and performance issues. The researchers have developed a prototype implementation atop Ethereum, demonstrating scalability without the harmful effects of naïve scaling up of computation time. The solution will enable the integration of applications using heavy computation, including AI/ML and privacy-preserving cryptography, and benefit industries such as Blockchain, Software, Finance, and Banking. A patent has been filed, and the security analysis is ongoing.

- **Indigenous 5G Packet Core**

Professor Mythili Vutukuru from the Computer Science and Engineering department has developed an indigenous 5G core network that is part of and fully integrated with the more extensive end-to-end indigenous 5G/5Gi network in India. The solution is standards-compliant, cloud-native, high-performance, scalable, and follows industry best practices. This solution is a starting point for indigenously developed 5G products in India, allowing the country to mature the product ecosystem and become a technology producer rather than a consumer. Developing Indian solutions to 5G will enable the focus on India-specific problems and push the nation's ideas into future standards. Having an Indian, secure 5G core is especially important for networks like defence and police. The solution will benefit the Telecom and ICT industries, and the 5G core network will soon be ready for campus deployment.

- **Best Women-led Startup and Best Startup in Life Sciences**

Prof. Manoj Gopalkrishnan - the CEO of Algorithmic Biologics Pvt. Ltd. and an Associate Professor at the Electrical Engineering department of the institute - and his team at Algorithmic Biologics won the Startup India 2022 Award for Best Startup in Life Sciences as well as the Best Women-led Startup, making it the only startup that won in both categories. Their research project 'Compressed Sensing Algorithms for Group Testing of Pooled Samples' was funded by the institute's Wadhwani Research Center for Bioengineering (WRCB) IIT Bombay's intramural funding support in 2020, under COVID special call for proposals.

- **DetFroSS: Frozen Soil Shear Strength Estimator**

Professor D. N. Singh from the Civil Engineering department has developed the DetFroSS set-up (patent filed) to determine the shear strength of frozen soil at different temperatures. The invention is designed to analyze soil's shear strength, density, saturation, and moisture, which are essential in designing building foundations in different temperature zones. The Frozen Soil Shear Strength Estimator quickly and economically determines the shear strength of frozen soil and volumetric deformation that could impact the foundation during construction. The setup will be helpful in executing infrastructure development, mainly in Northeast India, where the soil temperature is up to 25°C. The technology will benefit ecologists and engineers to design buildings and infrastructure that can withstand harsh environmental circumstances, promoting the country's growth and development.

- **ThermCon-P: Thermal Consolidometer With Pore-Pressure Profiling**

Prof. D. N. Singh has also developed a thermal Consolidometer With Pore-Pressure Profiling (patent filed), which helps to estimate the thermo-mechanically induced consolidation of soils. This setup is unique because it accurately records pore-water pressure and suction that builds up in the soil specimen. The data and analysis obtained through this setup enable the precise computation of soil consolidation under various thermo-mechanical testing conditions, including the simulation of various thermo-mechanical stress loadings on the soil specimen. The present study setup will help construct structures like underground utilities, satellite launching pads, geothermal energy utilisation assemblies, nuclear waste disposal structures, pipelines, etc. Currently, the setup is calibrated and has obtained promising preliminary results.

- **National Centre of Excellence in Carbon Capture and Utilization**

IIT Bombay established the National Centre of Excellence in Carbon Capture and Utilization (NCoE-CCU) with the support of the Department of Science and Technology, Government of India. The NCoE-CCU will serve as a centre for multidisciplinary, long-term research, design development, team-building, and capacity-building. In addition to creating innovative techniques for raising the technology readiness levels in CCU, it will establish milestones and lead science and technology activities for industry-oriented CCU innovation in India, thus quickly converting R&D into techniques for capturing and using carbon. Alongside, it will also study approaches that will improve hydrocarbon recovery as a co-benefit, as well as the conversion of captured carbon dioxide to chemicals, CO₂ transport, compression, and usage.

- **Compliant Mechanisms for Ultra-Precision Linear and Rotary Motion Control**

Professor Prasanna Gandhi from the Mechanical Engineering department has developed a new technology that uses compliant mechanisms for ultra-high-precision positioning of critical optical systems. Traditional ball-guided stages are limited by

friction, piezo actuators have limited travel range, and magnetic levitation-based stages are expensive and energy-intensive. Compliant mechanisms are friction-free, require no lubricants, have no wear and tear, and promise long life with high reliability. The technology has been successfully tested in the laboratory, and a part of it has been licensed to two companies, with one receiving a prestigious FIE Foundation award at IMTEX 2019. A startup called Flexmotion Technologies has been founded to bring the technology to the market. A patent has been filed and the technology has the potential to impact several industries, including defence, aerospace, robotics, manufacturing, and semiconductor equipment.

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- **Selective hydrogenation using 3d-transition metals**

- Professors Leela Srinivas Panchakarla and Santosh J. Gharpure from the Chemistry department have developed a cost-effective and environmentally-friendly solution for selective hydrogenation. Selective hydrogenation is a crucial reaction in organic synthetic chemistry used in the production of intermediates for industries like pharmaceuticals, fine chemicals, and biomass valorization processes. Traditional precious-metal-based catalysts for hydrogenation are inefficient, expensive, and hazardous. The Cobalt-Nickel bimetallic nanocatalyst supported on g-C₃N₄ developed by the professors can selectively hydrogenate α,β -unsaturated carbonyl compounds and nitroarenes under mild conditions. The solution reduces the cost of the product, minimizes the usage of heavy metals, and eliminates disposal issues. The technology is still in the proof-of-concept stage, and the patent is in progress.

- **Smart IoT Device**

- Prof. Asim Tewari from the Mechanical Engineering department has developed a Smart IoT Device that can sense various physical quantities and perform onboard analytics to detect machinery faults before they cause considerable damage. The device is integrated with sensors such as accelerometer, gyroscope, AC-current, AC-voltage, temperature, humidity, light/IR light, and magnetic field sensors. While similar IoT devices are available in the market, this device's continuous data sensing and onboard analytics combined with cloud AI give it an edge. The Smart IoT Device has immense applications in the smart manufacturing sector, where it can help detect machinery faults and predict their failure, leading to cost savings and improved machine uptime. A limited set of devices has already been demonstrated in the actual shop floor environment.

- **Programmable Virtual Communication Hub: Pro-H 1000**

Professor Ashwin Gumaste from the Computer Science and Engineering department has developed the Pro-H X000 series, a portfolio of programmable routing platforms that offer industry-leading performance, security, port density, and system capacity. The system has a benchmark for low latency, low power consumption, and long-distance

ranges, as well as providing sub-10 microsecond port-to-port latency across IPv6. The unique cloud network management system (NMS) and network analytics embedded in the NMS make this system an important tool for digital transformation in industries such as telecom, defence and security. The Pro-H 1000 SDN router can also be customized for use cases such as bandwidth calendaring, packet analytics, and security group creation. This system can provide a paradigm shift in routing and switching and offers key benefits to digital transformation for service providers, cloud operators and enterprises.

- **Competitions organised by Abhyuday, the social body of IIT Bombay**

Abhyuday's virtual science exhibition competition, the 'Social Innovation Challenge', witnessed participation from over 50 schools across India. Its purpose was to encourage children to devise innovative solutions to environmental problems through science models. Another exciting event was the 'Action Plan', a social entrepreneurship competition encouraging students to apply their academic learnings to various social issues to develop novel and feasible solutions.

- **Tinkerer's Lab**

IIT Bombay's class of 1975 launched a legacy initiative called the Tinkerer's Lab. Its goal is to encourage pupils to engage in exploration, creativity, and innovation. The Tinkerer's Laboratory (TL), which was inaugurated on March 19, 2014, offers a conducive physical setting for young engineering students to get hands-on and give form and expression to their inventive and imaginative natures. It also serves as a business incubator for students' aspirations, allowing them to unleash their creativity, foster their sense of adventure, and devise impactful products. It consists of electrical, mechanical, and software tools, and is equipped with a broad variety of equipment including grinders, CNC mills, soldering irons, 3D printers, hand drills, and CNC lathes. It also has access to the most cutting-edge technologies for prototyping. In January 2023, a workshop to train 80+ students and help them gain hands-on experience on all the machinery at the lab was organised by the TL team.

SDG 10: Reduced Inequalities

- **GRA: Group for Rural Activities**

The institute's GRA is a voluntary student group aiming to provide exposure to rural India to the diverse community of the institute. Their major activities include the GRA semester-end camp, in-semester visits to NGOs, and social internships. Furthermore, they organise numerous camps in which students visit pioneers of grassroots-level development across India to learn and understand how NGOs have contributed to the betterment of the villagers' lives. Moreover, they collect the discarded bicycles - left behind by the graduates - and distribute them among needy children in rural and tribal areas. The group has also organised multiple talks by prominent social workers. Their

'Teach Me' program provides a learning platform for underprivileged students from Phule Nagar, a slum area near the institute.

- **Supporting and uplifting students from historically marginalised backgrounds**

IIT Bombay offers affirmative action and reserves 27%, 15%, and 7.5% admissions for students belonging to Other Backward Classes - Non-Creamy Layer (OBC-NCL), Scheduled Castes (SCs) and Scheduled Tribes (STs) respectively. Additionally, the institute provides them with several other support measures to foster a better learning environment, for instance, exemption from the payment of tuition fees and hostel room rent, free messing, and pocket allowances. The institute also has an SC/ST student cell in place that mandates zero caste-based discrimination, addresses academic and non-academic issues faced by SC/ST students, and any complaints they may have, thereby helping them through every step of their journey.

- **Research at SJMSOM**

Prof. Varadraj Bapat, at the SJM School of Management, researches into financial inclusion, bankruptcy prediction, family business, and the Indian business model, with an overarching focus on sustainability. In the past, he has worked on a project titled 'Sustainability of self-help groups for financial inclusion'. Currently, his research focuses on a plethora of issues - migration in Northeast India, women empowerment by promoting social entrepreneurship, financial inclusion and corporate social responsibility, and environmental sustainability in India.

SDG 11: Sustainable cities and communities

- **STP-on-Wheels: Decentralized Onsite Sewage Treatment Plant**

Prof. Anil Kumar Dikshit of the Environmental Science and Engineering Department is working on a novel and highly effective technology for on-site and mobile treatment of municipal wastewater. A tremendous amount of wastewater is produced in cities and transported over great distances to traditional centralized wastewater treatment systems (CWTs), which causes several operating issues. In reality, the CWTs are often unable to handle significant sewage loads. As a result, the wastewater is occasionally dumped into water bodies without being fully treated, which has a negative impact on the environment. Prof. Dikshit is developing a potable mechanism for the safe and complete treatment of wastewater at source. The suggested system treats the wastewater in three different ways: i) through a centrifuge, producing treated water and biogas for reuse; ii) through a combination of aerobic, facultative, and anaerobic treatment; and iii) through a pre-settling tank and mechanical aeration unit. The project's primary advantage is the decrease in the harmful environmental effects caused by wastewater that hasn't been fully or adequately treated.

- **Research on sustainable materials, infrastructure, and HVAC systems**

Prof. Chaaruchandra Korde, an Assistant Professor at CTARA (Center for Technological Advancement of Rural Areas), focuses on sustainable materials and infrastructure research with an emphasis on structures and construction. At present, his students are pursuing research on 'Bamboo structural elements for sustainable mass housing and infrastructure applications', 'Low-cost portable cold storages for marginal farmers to increase their vegetable holding capacities', 'Protected cultivation using a bamboo polyhouse for enabling cultivation during peak rains' and 'Launchable platforms with bamboo for flood Impact reduction on villagers'.

Prof. Gurubalan Annadurai is an Assistant Professor in the Department of Energy Science and Engineering whose research interests lie in the field of energy recovery exchangers, HVAC (heating, ventilation, and air conditioning) systems and desiccant humidity control dedicated to creating a sustainable built environment. Currently, his doctoral and graduate students are pursuing research in topics related to sustainable HVAC systems.

- **Ensuring sustainable living through awareness generation and competitions**

We at the Sustainability Cell are striving to make a positive impact on the campus population by organising events that encourage people to incorporate sustainability into daily decision-making. In cooperation with the Maharashtra Pollution Control Board, we hosted the 'Airathon Mumbai' competition (as a part of Run4netZero, a climate change awareness campaign) for modelling air quality. The challenge was to come up with creative technological interventions to solve the problem of Mumbai's declining air quality. Students from all around India responded enthusiastically to the event, which received 150+ entries, with the best presentations on air pollution modelling and campaigning receiving financial awards.

Moreover, we are critically evaluating the consumption habits, waste production, and administration of 14 hostels housing over 13000 students to determine which ones are the most sustainable. The first-ever 'Green Cup' will be awarded to the hostel that generates the smallest carbon footprint in terms of per capita food, water & electricity use, and waste generation. We are hopeful that increased awareness among students through the spirit of competition will gradually lead to a more sustainable campus.

- **Team Shunya's Net-Zero Housing**

Team SHUNYA, a student initiative, endeavours to build a net zero housing structure under three main domains, i.e., energy, water, and carbon footprint. They are participating in the US Solar Decathlon, hosted by the NREL (National Renewable Energy Laboratory) and have received a prize funding of \$50,000. The steel structure they are developing is filled using eco-boards composed of agro-waste. Coupled with

sustainable landscaping, the structure offsets more carbon than is emitted during its construction and lifetime. They achieve net zero using three methods, namely, Rainwater Harvesting, Grey Water Recycling and Water-Efficient Fixtures, which have helped them achieve an awe-inspiring 82% more water savings than a normal house, making the structure an IGBC Net Zero Water House.

Moreover, they ensure that the house is net positive energy, which means it produces more energy than it requires, transporting the rest to the grid. In addition to the improved technology being used, we are introducing home automation to bring the future to the present, which increases the efficient consumption of energy within the house, along with providing luxury and comfort to the occupants at a single tap. Moreover, Team Shunya hosted its first-ever online event fest – ‘Sustain’, as a part of the team’s social awareness program towards sustainable houses. The main aim and objective of the event were to spread awareness about sustainability, technology and the idea behind it. ‘Sustain’ was held from October 31, 2020, to November 1, 2020, with six different subsystem-specific webinars for the participants. Hundreds of students participated in the two-day online fest.

- **Activities of the Energy Club**

The Energy Club, IIT Bombay, offered courses on 'Global Strategy, International Relations, and Management' and 'Fundamentals of Energy: Foundation to a Sustainable Future' in collaboration with Career Cell as part of the Technical Summer School 2022. The first course provided college students with a broad overview of the role that global politics and international relations play in driving energy investment, strategies and businesses. The second course was a deep dive into the field of energy, covering aspects like renewable energy, electric vehicles, storage sustainability, and climate change. Moreover, the Energy Club also conducted a solution-focused debate among postgraduate students on the virtue of labelling investments in nuclear plants and natural gas projects under the 'European Green Deal' as ‘sustainable’. Additionally, they conducted two interactive sessions on Electric Vehicle Technology with Anupam Bhattacharjee of Tata Technologies and the practical know-how on using EVs in India with Sushil Reddy.

- **Research at CUSE**

Prof. Himanshu Burte is an Associate Professor at the Centre for Urban Science and Engineering (CUSE) whose research revolves around urban policy and access to housing and public spaces to make cities just, equitable and sustainable. Currently, his group is pursuing research on ‘Urban disasters and resilience’, ‘Implications of policy on non-notified slums’, and ‘Understanding compact form in contemporary Indian cities’.

Prof. Arnab Jana, also an Associate Professor at CUSE, works on sustainable urban infrastructure planning, urban informatics, and behaviour and choice modelling for

transport users in India. His group is researching 'Integrating mode selection patterns and travel behaviour of heterogeneous society in an Indian context', 'Optimising urban energy consumption through urban planning interventions' and 'Land acquisition, rehabilitation and resettlement policies'.

Prof. Pradip Kalbar, another Associate Professor at CUSE, explores issues with urban environmental infrastructure from a life-cycle thinking and circular economy perspective. Currently, his group is pursuing research on 'Network performance-based adaptive design and operational strategies for water supply systems', 'Demolition waste management: circular economy approach', 'Resource recovery from sewage', and 'Liquid waste management in urban areas using natural treatment systems'.

SDG 12: Sustainable Production & Consumption

- **Zero Waste at IIT Bombay's 2022 Half Marathon**

The institute's half marathon - which welcomes over 4000 participants every year - once generated 156kg of single-use plastic water bottles as waste, when around 13,000 polyethylene terephthalate (PET) bottles were used to quench the thirst of the runners at the hydration stations along the course. These PET bottles were replaced with earthen cups at the Fitizen Half Marathon. A group of artisans called Asian Tigers offered to make 13,000 kulhads (glasses made of baked clay) as an alternative to plastic water bottles. A team of 100 volunteers composed of students, faculty members, and residents from the campus participated in transportation, cleaning, filling and post-use segregation of the kulhads. Several hydration points were set up along the running route, where volunteers kept filling the kulhads and handing those to runners. Some runners went a step ahead and decided to pour the water directly into their mouths (a practice common in India) instead of sipping through the kulhad so that it can be used multiple times by refilling. The response to the first half-marathon in India free of plastic bottles was overwhelmingly positive. Dr. Ankita Asthana, the marketing head of Fitizen, said, "This marathon would be an inspiration to conduct all other marathons in a plastic-free manner." Participants praised the uniqueness of the initiative and remarked that this had been the first marathon with such an initiative, and they were glad to be a part of it. Several participants took their used kulhads home as souvenirs.

- **Daksh: A Household-cum-Community Scale Composter**

Prof. Anil Kumar Dikshit of the Environmental Science and Engineering Department is creating efficient mechanisms for the daily production of useful compost. This will solve the problem of ineffective and improper solid waste management (SWM) that has impacted many lives due to the rise in garbage on the streets, pollution, offensive odor,

subpar waste disposal facilities, and related health issues. Composting devices currently available in the market are not designed to encourage rapid composting in the least amount of space. The research team is devising a solution that will provide an affordable design to promote ideal environmental conditions for the biodegradation of household organics. It provides daily waste storage, composting, and leachate collection. The product is currently in its testing and prototyping stage. Further research is being done on the reactor's design elements, and early work has been done on composters ranging in size from homes to communities. The household/community composter helps the Government's Swachh Bharat Abhiyaan (Clean India Mission) by reducing the load on landfills.

- **Activities of Team Zero Waste, a student-led initiative**

Team Zero Waste, Sustainability Cell, the Public Health Office (PHO), and the Hostel Council collaborated to join the residential recycling group. The goal was to start a new segregation method on campus, using three categories: food waste, landfill waste, and burnable waste. The project strategically planned the placement, shape, structure, and number of bins for effective segregation and significant change. Various NGOs conducted speaker sessions to educate people on the recycling process. An awareness session called the Plastic Recyclothion was organized in collaboration with Project Mumbai. Students could donate their plastic waste, and volunteers from the team picked it up from the closest drop location on campus. The Recycle-Resell-Reuse Facebook group was launched to promote recycling and reusing. The idea was to provide an online platform to solve the problem of ineffective communication and lack of a platform between buyers and givers.

Team Zero Waste also held an interactive session with the students of the Campus KV School to discuss the importance of proper waste management. They emphasized that a more effective waste management system, prioritising waste reduction and segregation at the source, can only be achieved through the active participation of all campus residents, including faculty, staff, IIT Bombay students, school students, and other residents. By involving everyone in the process, they aimed to create a more sustainable and eco-friendly environment on campus.

Moreover, the team also conducted various campaigns and projects like Greens to Gaushala, Nature Pays Back and Zero Waste Week to generate awareness for waste management, emphasizing on the facts and figures, problems from social, environmental and economic point of view, and solutions in terms of regulations and practices.

- **Automated and flow-controllable systems**

Buildings that have recently been built in IIT Bombay have sensor-based lighting and water tap systems, which cut down on the wastage of water and electricity. Examples include the Environmental Science and Engineering Department (ESED) building, the Victor Menezes Convention Center (VMCC), and the Lecture Hall Complex (LHC),

among others. The institute's annual report outlines that future buildings will be outfitted with similar technologies, which will go a long way in reducing their environmental burden.

- **Research into corporate environmental strategy**

Prof. Kirankumar S. Momaya is a Professor at the SJM School of Management whose research interests revolve around institute social sustainability, organisational competitiveness, and the natural environment. In the past, his research focused on organisational innovations for sustainable growth and grassroots innovations improving health. Currently, Prof. Momaya's group is working on corporate environmental strategy for improved competitiveness and environmentally sustainable practices in firms. Additionally, he is a part of the semi-annual CSR initiative of the Group of Competitiveness at IITB, called the 'Biathlon' which focuses on community fitness and maintaining the greenery of the campus.

- **Waste management facilities**

IIT Bombay has implemented an effective waste management system that diverts most of the waste from reaching landfills. The system employs a biogas plant and composting facilities to treat wet and dry biodegradable waste. The biogas plant uses anaerobic digestion to treat wet waste and produces methane and other gases that are used as a form of energy. The methane obtained is used as fuel for cooking in hostel messes, and the slurry is used as fertiliser for horticulture. Composting facilities treat dry biodegradable waste such as leaves and wet waste from the residential sector. The process involves adding the waste in a ventilated chamber or a pit and leaving it undisturbed for a specified period with tilling or mixing in known intervals. The facilities receive waste from staff quarters and residential buildings, except hostels. The process requires the maintenance of an adequate environment and is not carried out during the monsoon season. The resulting compost is used as fertiliser for horticulture within the campus.

- **Plastic Recycling Drive at the campus school**

A plastic recycling drive was recently organized at the IIT Bombay campus KV school, focusing on multilayer wrappers that were previously difficult to recycle. To encourage student participation, a week-long challenge was introduced in which secondary school students were asked to segregate multilayer plastics in PET containers, create eco-friendly posters, and write essays on plastic recycling. Team Zero Waste also conducted a pilot run in collaboration with NSS, for plastic segregation in hostel H15-B. The initiative aimed to raise awareness about the importance of responsible waste management practices and the role of plastic recycling in creating a more sustainable environment.

- **Cloth Collection Campaign by the National Service Scheme (NSS)**

NSS, IIT Bombay, through its 'Cloth Collection Campaign', collects old clothes from students and other campus residents that can still be used and makes them available to the needy free of cost through different NGOs working for the upliftment of the poor and underprivileged. They have collected 3-4 tons of waste cloth each year for the past 5-6 years, which accounts for roughly 15 tons of waste. This initiative has helped them save a whopping 375 tons¹ of CO₂!

SDG 13: Climate Action

- **Interdisciplinary Program in Climate Studies (IDPCS)**

The IDPCS was initiated at IIT Bombay under the Indian Government's Department of Science and Technology - Centre of Excellence in Climate Studies (DST-CoECS) project in 2012 as one of India's first doctoral programs focussing on research related to climate change. Currently, 31 faculty members drawn from 12 departments across IIT Bombay are associated with IDPCS, who increasingly apply a depth of expertise in their fields to address the complexity of climate change. They apply their expertise to interdisciplinary climate problems, transcending traditional academic boundaries, and contributing to traditional core climate and environmental sciences, interdisciplinary technology development, climate economics, and social sciences. The IDPCS also organises panel discussions and participates in climate science competitions. The initiatives from the climate studies department are designed to provide students and researchers with a comprehensive understanding of the Earth's climate and the current state of climate research, preparing them for careers in climate science or related fields.

- **Research at the HSS Department**

Prof. K. Narayanan of the Department of Humanities and Social Sciences (HSS) works on energy economics, industrial economics, industry-environment linkages and the socio-economic impacts of climate change. His research has focussed on the effects of climate change on the household economy and the adaptive response of agricultural households in India.

Prof. D. Parthasarathy, also at the HSS Department, explores urban studies, vulnerability and adaptation to climate change, sustainable transformation, and disaster studies. His primary research areas include vulnerability and adaptation assessment tools, gender-based climate justice, and agricultural adaptation strategies.

- **Grow Green Gold by Abhyuday**

Abhyuday, the social body of IIT Bombay, conducted a nationwide plantation drive, called 'Grow Green Gold', encouraging over 300 people to plant trees in a single day and inspiring many more to pledge to plant trees to combat deforestation and climate change. Additionally, they organised a case study competition to promote fundamental consciousness about climate change and ways to tackle it among the youth.

- **Electric buggies for in-campus transport**

The institute has recently introduced a fleet of electric buggies on campus to transport students. These buggies are environmentally friendly and provide a convenient way for students to get around the campus. The buggies are powered by electric motors and are equipped with GPS and real-time tracking systems. The buggies are available at reasonable prices to students, staff, and faculty members. The use of electric buggies is in line with the institute's commitment to sustainability and reducing its environmental impact.

SDG 14: Life Under Water

- **Abhyuday's Powai Lake and Versova Beach Clean-Up**

Abhyuday, the social body of IIT Bombay, conducts regular clean-up drives at Powai Lake and Versova Beach. The goal of these clean-up drives is to preserve the natural beauty of these areas and protect the local environment. The initiative aims to raise awareness among the institute community and the general public about the importance of preserving natural resources and keeping our environment clean. The clean-up drives are conducted regularly and involve a large number of volunteers who work together to collect and dispose of trash and debris. In the Powai Lake Mega cleanup, 800+ volunteers participated in collecting over 2.5 tonnes of solid waste along the 2km bank of the lake. The initiative also focuses on educating people about the impact of pollution and encouraging them to adopt sustainable practices to reduce their environmental footprint.

- **Eco-friendly Ganesha idols by the National Service Scheme (NSS)**

The 'Eco-Friendly Ganesha' initiative of NSS, IIT Bombay, invites the entire institute, and even outsiders, to make Ganesha idols from eco-friendly materials like clay. This initiative has been running in the institute for the past 20 years in collaboration with the Industrial Design Centre (IDC). Every year 100+ students and 200+ plus faculty & outsiders participate in this activity making around 50-60 idols. This initiative helps reduce ~300 kilograms of PoP waste and toxic paint materials (like lead & cadmium),

helping maintain the ecology of the nearby water bodies as the idols are immersed into them during festivities.

SDG 15: Life on Land

- **Activities of the National Service Scheme (NSS)**

The National Service Scheme (NSS) at the institute promotes sustainability and environmental conservation through various initiatives. The successful tree plantation drive last year resulted in the planting of over 200 saplings and 45+ small native plants, along with 400 bottle-made plantation models and 100 bottle-made bird feeders. The NSS also organized a Green Diwali campaign and a paper recycling drive to reduce waste on campus. The drives and campaigns help to improve air quality, conserve biodiversity and create a more conducive environment for study and research. These initiatives aim to create a more eco-friendly campus and raise awareness among students about the importance of sustainability and environmental conservation.

- **Talk about the Healing Himalayas Foundation**

IIT Bombay's Team Zero Waste organised a session for students with Pradeep Sangwan, the founder of Healing Himalayas Foundation (HHF). HHF has taken up the onerous task of cleaning out the Himalayas to make the picturesque spot more tourist friendly. The foundation has cleared almost 900 metric tonnes of non-biodegradable waste from the foothills of the Himalayas to date. The session impressed upon the students the importance of protecting the environment, local communities, and wildlife.

SDG 16: Peace, Justice and Strong institutions

- **Activities of Abhyuday - the social body of IIT Bombay**

Abhyuday organised 'Sandesh to Soldiers', an initiative wherein numerous people wrote about their feelings of love and gratitude towards our soldiers and policemen as a token of appreciation for their undeterred courage, and the heroic sacrifices they have made for society and the country. Moreover, Abhyuday conducted a PIL (Public Interest Litigation) drafting competition to increase student awareness about laws and policies pertaining to social issues. The winning PILs are being forwarded to and filed in the Honorable Supreme Court of India and various High Courts.

- **Multi-resolution ML for touchless biometric applications**

Professor Vikram M. Gadre from the Electrical Engineering department is leading research on touchless biometric identification and verification techniques based on multiresolution machine learning/deep learning techniques. The research has the potential to make biometric authentication more robust, particularly in the pandemic era. The use of touchless biometrics in security applications such as airports, border control, corporate offices, and banks, can indirectly contribute to the safety and well-being of people by detecting fraud and preventing security breaches. The researchers have developed a prototype mobile app for touchless fingerprint recognition and classification, and are integrating the investigations of biometric systems based on the ear and iris into a software package. The development of touchless biometric solutions is user-friendly and hygienic, and can be used in law enforcement and the healthcare industry. Overall, the research has the potential to provide a national resource for use in UIDAI, police, defense establishments, security applications, and use in other corporate/government establishments. Using AI/ML/Blockchain technologies (including quantum computing) to further enhance the accuracy and security of touchless biometric identification and verification techniques, the project is a promising solution for the security, defense, banking, airports, and healthcare industries.

- **Centre for Policy Studies**

The Centre for Policy Studies (CPS) at the institute conducts interdisciplinary and applied research on policy issues such as sustainable development, public health, education, and governance. CPS research teams consist of faculty, research staff, and graduate students who produce high-quality research relevant to policymakers. The CPS collaborates with other research centres like the Humanities and Social Science (HSS) Department, the school of management (SJMSOM), the Industrial Design Centre (IDC), and the Centre for Technology Alternatives for Rural Areas (CTARA) to conduct research on various policy issues. The CPS regularly organizes workshops, seminars, and conferences to disseminate the findings of its research to the broader community and engage in policy discourse.

- **Vajra - Threat Detection and Response in Linux End Points**

Prof. Manjesh K. Hanawal and his team at the Industrial Engineering and Operations Research centre have developed an endpoint detection and response (EDR) tool, 'Vajra' for Linux systems with container security support. This is a unique solution as most EDR solutions only cater to Windows systems. It allows real-time detection of lateral movements and privilege escalation, enabling system administrators to respond quickly to attacks. The team plans to provide continuous support and update the tool to detect and prevent new malware. This tool helps build a safer digital society by ensuring privacy and security for all users of digital services. The solution is relevant to the cybersecurity and information and communication technology industries. The need for an integrated Endpoint Security solution that can quickly detect, analyze, block, and

contain attacks in progress is essential, especially with the rise of fileless malware attacks that are difficult to detect using standard methods.

SDG 17: Partnerships for the Goals

- **International Energy Symposium**

IIT Bombay's Energy Club hosted the International Energy Symposium in collaboration with the energy clubs of TU Delft Netherlands and NTU Singapore on the 1st and 2nd of June, 2022. The symposium brought together people from different sectors to discuss pertinent technical and social aspects of green energy. Panel discussions in the event shed light on specialized topics like green hydrogen, smart grids, and green economics, as well as social issues like improving the inclusivity of women and marginalized groups in energy policy discussions.

- **MoU signed with Maharashtra University of Health Sciences**

IIT Bombay and Maharashtra University of Health Sciences (MUHS) signed a Memorandum of Understanding (MoU) to enhance collaboration between the two institutions. The MoU enables the sharing of resources for academic and research purposes in the diverse areas of mutual interest, particularly in the field of health sciences. This collaboration aims to bring together the technological advancements of IIT Bombay and the expertise in health sciences of MUHS, providing opportunities for interdisciplinary research and development in the field of health sciences. The MoU is expected to encourage more people to explore digital health and provide valuable services that will be compatible with the future of healthcare.

- **MoU with the Technical University of Denmark**

During the Danish Royal Visit to India, the Technical University of Denmark and IIT Bombay exchanged a Memorandum of Understanding and Confidentiality Agreement to establish a Strategic Partnership. This MoU is anticipated to foster collaboration among leading researchers from both India and Denmark, who are committed to addressing global challenges. The agreement signifies a long-term partnership between these distinguished institutions within the areas of science, innovation, and education. These areas include, but are not limited to, waste water treatment and the impact of climate change. The significance of this partnership lies in the mutual recognition of the importance of research, education, and innovation for achieving sustainable development goals. This MoU represents a strong foundation for continued collaboration between these institutions and an opportunity to advance global progress through joint efforts.

- **SINE's Partnerships in infrastructure, healthcare, and finance**

IIT Bombay's Society of Innovation and Entrepreneurship (SINE), fosters innovation and research activities, ultimately helping convert them to entrepreneurial ventures. SINE is actively involved in establishing memorandums and partnerships with many companies and municipal corporations for the mutual flow of technologies and practical instances. Some organisations SINE has successfully established partnerships with are the Brihanmumbai Municipal Corporation's (BMC's) Society for Mumbai Incubation Lab to Entrepreneurship (SMILE) for infrastructure, the Consortium of Accredited Healthcare Organizations (CAHO) for healthcare-related startups, and the Reserve Bank Innovation Hub (RBIH) to aid innovation in the financial sector.

- **IITB-Monash Research Academy**

The IITB-Monash Research Academy, a collaboration between IIT Bombay and Monash University, Australia, fosters a solutions-driven approach to solving global problems by providing PhD training on a sizable scale. It is taking the lead in resolving some of the biggest problems of our time, thanks to robust industrial participation. With its joint PhD program, graduate students from India work with both IITB and Monash supervisors and business partners. Students get a joint Doctorate degree from IITB and Monash University after spending at least a year in Australia, thus offering a genuinely global education. The IITB-Monash Research Academy's outputs are focused on having an impact and utilising the combined assets of two extremely talented organisations.

- **Partnership With Washington University in St. Louis**

IIT Bombay and Washington University in St. Louis collaborated to lead to India's first and the only joint EMBA degree program for working professionals and entrepreneurs. The partnership rests on the common belief that the continuous evolution and transformation of executive education will be the most important catalyst enabling strategic transitions in the 21st century. Keeping in line with this thought, the joint Executive Master of Business Administration (EMBA) degree focuses on making leadership succeed and allowing change-management thrive. The course aims to make an individual a more confident, competent, and consistent professional or entrepreneur. It will make them future-ready, transition-ready, and eager to embrace every challenge or opportunity with grace, intelligence, and intuition.

Moreover, researchers from the McKelvey School of Engineering at Washington University in St. Louis have been collaborating with IIT Bombay to establish the Aerosol and Air Quality Research Shared Facility in Mumbai, along with running a joint master's degree program. The research facility joined the Surface Particle Matter Network (SPARTAN), a global particulate matter network, which connects satellite mapping and particulate matter sensor deployment. The partnership also brought to the IIT Bombay campus high-tech instrumentation for fundamental research in aerosol science and engineering, as well as new equipment for on-the-ground air quality monitoring and characterization.

- **Sponsors and partners for student bodies promoting sustainability**

The institute has several student bodies like the Sustainability Cell, Abhyuday, Team Shunya, Team Zero Waste and NSS, that are dedicated to promoting sustainable activities on campus. These teams work on various initiatives, such as promoting renewable energy, reducing waste, and increasing recycling. These teams have secured sponsorships and support from various organizations, including government agencies, private companies, and non-profit organizations. The sponsorships provide the teams with the financial and logistical support they need to carry out their activities and make a meaningful impact on the institute's sustainability efforts. The teams work closely with the sponsors to ensure that their activities align with the sponsor's sustainability goals and objectives. Current and past sponsors include Indian Oil, Larsen & Toubro, Pepperfry, Bosch, and many others.

- **Industry Collaborations**

IIT Bombay with its persistent efforts has maintained strong alliances with several renowned national and international companies through joint research and development activities and consultancy services. The AY 2021-22 saw the signing of more than 250 agreements for research collaboration, establishment of consortia/centers of excellence, licensing, non-disclosure agreements, intellectual properties transfer, student sponsorships, endowment and many more with businesses, organizations, universities, and governments. Nearly every societal sector and a wide range of businesses, including consultancy, electronics, pharmaceuticals, aerospace, aviation, the power sector, FMCG, automotives, chemicals, IT & software, government institutions and others, are involved in the partnerships. A few notable ones are L&T, Google, Pfizer, Vedanta, Honeywell, Amazon, Mahindra, Mercedes Benz, Godrej, Adani Power, and Sony, among others.

- **The Asian Universities Alliance**

IIT Bombay is a part of the Asian Universities Alliance (AUA), which focuses on various aspects of international collaboration, including building a more sustainable future. The AUA comprises 15 leading universities from 14 countries in Asia, and aims to create a platform for academic collaboration and exchange, promote mobility of students, faculty, and staff, and foster research and innovation in the region. IIT Bombay's participation in such alliances further reinforces its commitment to promoting international cooperation and contributing towards global sustainability efforts through collaborative research, academic exchange, and joint initiatives.

7. Spotlight on: Biodiversity

Much of this section has been borrowed from the 2020 Biodiversity Report by InSight, IIT Bombay's student media body. We would like to thank InSight and Ganesh Chelluboyina (batch of 2020) for the same.

The green, lakeside IIT Bombay campus is a biodiversity hotspot with at least 843 species of flora and fauna (WWF, 2008) living in its diverse, niche ecosystems. This rich biodiversity stems

from an overlap of the lake, wetland, hill-slope, grassland and forest ecosystems on the campus. It is important to note that the figure of 843 species could be an underestimation as the WWF report was “limited due to time and survey constraints”.

Core Biodiversity Hotspots



Green areas on the periphery represent the vegetated sector, and the yellow area represents the campus' urban sector, with abundant exotic, non-native trees lining the streets.

1. Soneri Baug

- Most accessible part of the campus stretching from Jal Vihar Guesthouse to Hostel 17
- Veritable hotspot for wildlife and popular among birdwatchers
- Major issues – Spillover effect from the construction of Padma Vihar Guest House and Hostel 17 along with its accessories, dumping of garbage and alcohol bottles, and rehabilitation of stray cows

2. Kol Dongri

- Large, unfragmented, and thickly-wooded forest
- Paradise for birdwatching, home to the indigenous ruddy mongoose
- Major issues – Construction of Hostels 12, 13 and 14 by clearing the forest, construction workers' colony, indiscriminate waste dumping, and poaching activities

3. SAMEER Hill

- Vegetated region with a rich slope ecosystem
- Undisturbed by development due to unfavourable topography
- Major issues – proposed construction of a solar energy park, waste dumping at its periphery

4. **Peru Baug and in-between**

- Next to Vihar Lake, has very rich biodiversity
- Has been largely untouched due to restricted access
- Region in between Peru Baug and Soneri Baug used to have thick forests until Hostels 15, 16, and 18 were constructed

Institutional Framework for Biodiversity Conservation

- The Green Campus Initiative (GCI) setup in 2010 under the office of Dean Infrastructure Planning and Support (IPS) was active in the first part of 2010s and had come up with a 31-point action plan to conserve campus ecology, energy, and water.
- The Strategic Plan Document was prepared in 2017, with preservation of greenery in campus being a major goal; accordingly, a five-year green master plan was proposed
- The 46th and 47th meetings of the council of IITs in 2013 recommended balanced infrastructure growth with environment conservation, and sustainable growth by establishment of Green Offices, creation of master plans for green infrastructure, and conducting regular 'green audits'.

Student Activities

IIT Bombay students have always been conscientious about their role in protecting the environment. Look at some student activities that took place in the '70s and '80s!

- **Formation of the Wildlife Club in 1977** in response to protests against road construction through Soneri Baug in 1976. Major activities – tree planting, birdwatching outings, national park visits, wildlife films, and exhibitions -conducted
- **Establishment of 'The Drongo'**, an environmentally focussed newsletter, in 1989 after another major resistance to tree cutting on campus. Its purpose was to foster a community of concerned individuals, and spread information about conservation efforts taken by the students.

Q. Food for thought: The most major challenge IIT Bombay faces in the near future is the ever-increasing student numbers and required infrastructure facilities. What can be done about it?

A.

- Going vertical for new constructions
- Redevelopment and remodeling of old hostels
- Space efficiency in new academic buildings
- Native over exotic plants on campus to support fauna
- Improved administrative policies
- Green Action Plan in consultation with campus community
- Revival of the Green Office
- Alumni Support for conservation projects
- Wildlife Club 2.0

8. The Team

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